



Vidyavardhini's College of Engineering and Technology, Vasai

Department of Computer Science & Engineering (Data Science)

EXPERIMENT ASSESSMENT

ACADEMIC YEAR 2025-26

Course: Introduction to Web Technology

Course code: 2344211

Year: SE SEM: IV

Experiment No. 8
Aim: Building a Simple Counter App using React Hooks. Use React Hooks (useState, useEffect) to manage state and side effects.
Name: Hetvi Patel
Roll Number: 56
Date of Performance:
Date of Submission:

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission.	10	
Total	20	

Checked by

Name of Faculty :

Signature :

Date



Experiment 8

Title

Building a Simple Counter App using React Hooks. Use React Hooks (useState, useEffect) to manage state and side effects.

Theory:

React is a popular JavaScript library used for building interactive user interfaces. Traditionally, React applications used **class components** to manage state and lifecycle methods. With the introduction of **React Hooks**, functional components can now manage state and side effects more efficiently.

1. React Hooks

Hooks are special functions in React that allow developers to “hook into” React features such as state and lifecycle methods from functional components. Hooks improve code readability, reusability, and reduce complexity.

2. useState Hook

The useState hook is used to manage **state** in functional components. State refers to data that can change over time and affects what is rendered on the screen.

In a counter application, the counter value is stored as state. Whenever the state changes, React automatically re-renders the component to reflect the updated value.

3. useEffect Hook

The useEffect hook is used to handle **side effects** in functional components. Side effects include:

- Updating the document title
- Fetching data
- Logging values
- Running code after component render



In this experiment, `useEffect` is used to update the browser title whenever the counter value changes, demonstrating how side effects are handled in React.

4. Component Re-rendering

Whenever the state changes using `useState`, React re-renders the component. The `useEffect` hook runs after rendering, ensuring that side effects are synchronized with the component's state.

Algorithm / Procedure

1. Start the experiment.
2. Create a new React application using `create-react-app`.
3. Open the project folder in VS Code.
4. Create a functional component for the Counter App.
5. Initialize counter state using `useState`.
6. Add buttons to increment and decrement the counter value.
7. Use `useEffect` to update the document title when the counter changes.
8. Display the counter value dynamically on the webpage.
9. Run the React application in the browser.
10. Stop the experiment.

Program / Code:

Counter Component (App.js)

```
import React, { useState, useEffect } from "react";
```

```
function App() {  
  const [count, setCount] = useState(0);
```

```
  useEffect(() => {  
    document.title = `Count: ${count}`;
```



Output:

4



Thus, a simple counter application was successfully developed using React Hooks. The experiment demonstrates the use of `useState` for state management and `useEffect` for handling side effects. This approach simplifies component logic and improves code readability in modern React applications.



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Questions:

1. What are React Hooks?

React Hooks are special functions in React that let you use state and other features in functional components (e.g., `useState`, `useEffect`).

2. How does React re-render components?

React re-renders a component when its **state** or **props** change. It updates only the changed parts using a virtual DOM.

3. How does React re-render components?

React re-renders when there is a change in state or props, and then updates the UI efficiently using the virtual DOM.